CAN KIDNEY EXCHANGE BE IMPROVED BY ALLOWING MORE DONORS?

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MASTER'S THESIS

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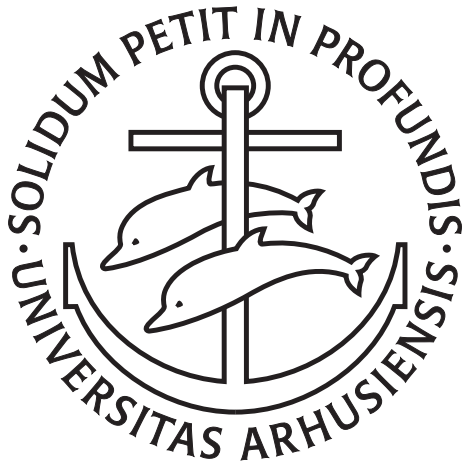
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Master's Thesis

Department of Computer Science
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ABSTRACT

This thesis investigates a generalized model of kidney exchange where each patient may be accompanied by up to two proxy donors. We formally define this multi-donor setting and analyze its computational and strategic implications. Through reductions from classical NP-complete problems, we establish the hardness of all variants and explore their approximability. We propose an effective approximation algorithm with a proven bound and analyze incentive properties under strategic donor selection, introducing a randomized mechanism to ensure truthfulness. Our experimental results demonstrate that the proposed model is not only theoretically robust but also practically viable. Optimal solutions can be obtained within minutes for realistic instance sizes, and the presence of multiple donors substantially improves match rates. These findings highlight the potential of multi-donor frameworks in enhancing the efficiency and equity of kidney exchange programs.

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1 INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Kidney transplantation is the most effective treatment for patients with end-stage renal disease. However, due to the severe shortage of deceased-donor organs, living kidney donation has become an increasingly important option. In many cases, a willing donor is biologically incompatible with their intended recipient. To overcome this, several types of kidney exchange (KE) donation programs have been studied and established in practice, allowing incompatible donor-patient pairs to exchange kidneys with other pairs in a mutually beneficial way.

One of the first attempts to model kidney exchange programs was based on the well-studied Top Trading Cycles model, which was later extended in [10] to include Top Trading Cycles and Chains. In these models, each patient-donor pair reports a preference ranking over all donors in the system. Each patient-donor pair is represented as a node in a directed graph, with an edge pointing to their most preferred available donor. The algorithm repeatedly identifies and removes cycles (groups of patient-donors who can exchange kidneys among themselves), updating preferences after each removal. This process continues until no cycles remain, ensuring that every exchange benefits all participants involved.

This class of models exemplifies a truthful mechanism, wherein each patient ultimately receives the best possible outcome according to their preferences. Despite this theoretical appeal, such models are not used in practice due to several limitations. First, each patient must report a preference ranking over all donors. In practice, KE programs involve thousands of donors, making it infeasible for patients to construct such detailed preference lists. Moreover, most patients are indifferent among compatible kidneys, as long as the match is viable. Second, the model permits cycles of unrestricted length. However, in practice, all transplants within a cycle must be performed simultaneously to avoid the risk of a donor backing out after their paired patient has already received a kidney. Consequently, KE programs typically restrict cycle lengths to two or three to ensure logistical feasibility and minimize the risk of failure. Lastly, the model focuses on truthfulness and not on optimizing the number of transplants. In practice, however, maximizing the number of transplants is often the primary objective of kidney exchange programs, as this directly translates to more lives saved.

These issues were addressed in [1], where the KE program is modeled as a graph in which *patient-donor pairs* are represented as vertices, and directed edges indicate compatibility between them. The goal is to find an exchange that maximizes the number of patients who receive a transplant, while restricting the length of cycles. When cycles are limited to length two, [1] shows at the beginning of Section 4 that the problem reduces to a maximum weight matching problem on a bipartite graph, which can be solved in polynomial time. Furthermore, when the restriction on the length of cycles is larger than 2, Theorem 1 in [1] proves that the associated decision problem becomes NP-complete.

In practice, some patients may have access to more than one incompatible but willing donor — for example, both a spouse and a sibling. Existing models can typically be extended to allow the use of *alternative donors*, with only one is ultimately selected in the solution. However, not many models are designed for many-for-one exchange, where *multiple donors* are willing to donate their kidneys simultaneously in exchange for a compatible kidney for their associated patient. Allowing this type of exchange introduces new possibilities for improving overall efficiency. At the same time, it raises new algorithmic and strategic questions. How should we model compatibility and exchange opportunities in this extended setting? How complex is it to find optimal allocations? Can we design mechanisms that are efficient and incentive-compatible?

One closely related idea permits the creation of *clubs*, groups of multiple patients and donors in which a donor is willing to donate if any patient from their club receives a transplant. A well-defined model building on and expanding this idea was introduced in [6], where, Theorem 1 shows that the decision problem associated with the uncapped generalized kidney exchange clubs problem is NP-complete.

However, while this model provides an expressive foundation, it does not fully address the challenges introduced by allowing multiple donors per patient under stricter feasibility and incentive constraints. This thesis builds on these ideas by proposing a refined formal model for kidney exchange with multiple donors per patient, analyzing its computational complexity, designing approximation algorithms, and studying strategic behavior under different mechanism designs.

1.2 CONTRIBUTIONS AND THESIS ORGANIZATION

This thesis explores a generalization of kidney exchange in which each patient may be associated with more than one donor.

Chapter 3 is about formalizing this multi-donor kidney exchange setting as a series of well-defined algorithmic problems. Notably, these problems are not specific to kidney exchange and may also

arise in other real-world domains involving structured matching or constrained resource allocation.

From a game-theoretic perspective, we further investigate the notion of truthfulness at Chapter 4. We formally define truthfulness in the context of multi-donor kidney exchange and prove that no deterministic algorithm can simultaneously achieve truthfulness and social welfare optimality. However, we demonstrate that by introducing randomization, certain deterministic algorithms can be transformed into truthful randomized mechanisms.

We then show that a series of well-defined algorithmic problems are NP-complete at Chapter 5 and further establish their approximability limits at Chapter 6. For one of the problems, we design a constant-factor approximation algorithm and formally prove its approximation guarantee.

Finally, we evaluate our approach on synthetic datasets at Chapter 7. Our experiments show that, in practice, optimal solutions for instances involving thousands of patients can be computed efficiently using integer programming. Furthermore, allowing a second donor per patient leads to approximately a 5% increase in the number of patients who receive transplants.

2 PRELIMINARIES

2.1 COMPUTATIONAL COMPLEXITY

In this section, we briefly review fundamental concepts from computational complexity theory that are relevant to the hardness results presented in this work. We focus on the notions of decision and optimization problems, the class NP, NP-completeness, and approximation hardness, which provide the theoretical foundation for analyzing the difficulty of algorithmic problems in kidney exchange. More detail can be found in [7].

Decision and Optimization Problems

Many computational problems can be framed in two ways: as a decision problem or as an optimization problem. A decision problem asks whether a solution meeting some criterion exists (e.g., "Is there a matching saving at least k patients?"), while an optimization problem asks for the best such solution (e.g., "What is the maximum number of patients that can be saved?").

Complexity theory typically studies decision problems, particularly those in the class NP.

The Class NP and NP-Completeness

A decision problem is said to be in NP (nondeterministic polynomial time) if a proposed solution can be verified in polynomial time. A problem is NP-complete if it is both in NP and as hard as any other problem in NP, meaning that every problem in NP can be reduced to it in polynomial time. NP-complete problems are believed to be intractable in the worst case—that is, they are unlikely to have polynomial-time algorithms.

To prove that a problem is NP-complete, it is common to reduce from a known NP-complete problem, such as *3SAT*, *Hamiltonian Path*, or *Longest Simple Path*. In this thesis, we use such reductions to establish the hardness of several matching problems in the kidney exchange setting.

Approximation and Inapproximability

When exact solutions are computationally intractable, approximation algorithms provide a practical alternative. An algorithm is said to have an approximation ratio α if it always produces a solution whose value is within a factor α of the optimal. For example, a 1/2-approximation algorithm for maximizing the number of transplants would guarantee at least half the optimal number.

Some NP-hard problems admit good approximation algorithms, while others are hard to approximate within any constant factor unless P = NP. In our setting, we also investigate whether meaningful approximation guarantees can be achieved despite the underlying NP-completeness of the matching problems.

2.2 RELATED PROBLEMS

Problem 1 (Longest Path from Fixed Source)**.** *Given a directed graph* $G = (V, E)$*, a designated source vertex* $s \in V$*, and an integer* k*, decide whether there exists a simple path starting from* s *of length at least* k*.*

This problem is NP-complete in general directed graphs, by a standard reduction from the *Hamiltonian Path* problem [7].

Moreover, the *Longest Path from Fixed Source* problem is not only NP-complete but also hard to approximate. In general directed graphs, the solution cannot be polynomial time approximated within $n^{1-\varepsilon}$ for any $\varepsilon > 0$ unless $P = NP$ [3].

Problem 2 (2P2N-3SAT)**.** *Given a Boolean formula* ϕ *in conjunctive normal form (CNF), where each clause contains exactly three literals, and each variable appears in two positive literals and two negative literals throughout the formula, decide whether* ϕ *is satisfiable.*

This problem is a well-known restriction of the classical *3SAT* problem, and it remains NP-complete under the stated constraints. The restriction limits the number of times each variable can occur positively and negatively, which is useful for reductions that require bounded variable degrees. The NP-completeness of this variant was established in [2].

In addition to being NP-complete, the *2P2N-3SAT* problem remains APX-hard [2]. Its maximization variant, which seeks to satisfy as many clauses as possible, admits no polynomial-time approximation scheme (PTAS), unless $P = NP$.

Problem 3 (k-Set Packing)**.** *Given a finite universe* U *and a collection* S *of subsets of* U*, where each subset has cardinality at most* k*, find a maximum-size subcollection* $S' \subseteq S$ *such that all sets in* S' *are pairwise disjoint.*

This classical optimization problem is known to be NP-hard for any $k \geqslant 3$, and it remains hard to approximate. Cygan [4] presented an algorithm that, for any $\varepsilon > 0$, achieves a $\frac{3}{k+1+\varepsilon}$ approximation ratio. The algorithm runs in time polynomial in the number of sets and elements.

2.3 MECHANISM DESIGN AND INCENTIVE COMPATIBILITY

In this section, we briefly introduce key concepts from mechanism design theory that are relevant to our work. More details can be found in standard references such as [11].

Mechanism Design Basics

Mechanism design studies how to construct rules or systems (mechanisms) that lead rational agents to act in a desirable way, typically by reporting private information truthfully. In the context of kidney exchange, each participant (i.e., a group of patient and donors) may possess private information about their available donors or compatibility relationships. The goal is to design a mechanism that selects a socially desirable matching while incentivizing agents to report truthfully.

Formally, let $\mathcal{A}$ denote the set of all agents (e.g., patients), and let each agent $i \in \mathcal{A}$ have a private type θ_i (e.g., their available donors). A mechanism consists of:

- A message space S_i for each agent i, where $s_i \in S_i$ is the reported type (which may differ from the true type θ_i);
- An allocation function $f : S_1 \times \cdots \times S_n \to \mathcal{O}$, which maps reported types to an outcome (e.g., a matching of patients to donors).

In our setting, the allocation function (essentially the algorithm that takes reports as input and produces a matching as output) is the central object of interest. In most cases, the primary design goal is to maximize social welfare, typically measured by the number of patients who receive a kidney.

Incentive Compatibility and Truthfulness

A mechanism is said to be *truthful* or *dominant-strategy incentive compatible (DSIC)* or *strategyproof*, if reporting truthfully is always in each agent's best interest, regardless of what the other agents report. For-

mally, for all agents i, all true types θ_i, and all possible misreports $s_i' \in S_i$, we require:

$$u_i(f(\theta_i, s_{-i})) \geqslant u_i(f(s_i', s_{-i})),$$

where u_i is the utility function of agent i, and s_{-i} denotes the reports of all other agents.

Ideally, our allocation algorithm satisfies three goals: (i) truthfulness (DSIC), (ii) maximizing social welfare, and (iii) computational efficiency (i.e., running in polynomial time).

3 PROBLEM FORMULATION

This chapter introduces the formal model that underpins this thesis. We consider a generalization of the standard kidney exchange framework, where each patient may be associated with more than one donor. We begin by reviewing the conventional patient-donor pair model and then extend it to accommodate the setting with multiple donors per patient. We subsequently define a series of algorithmic problems that emerge in this multi-donor context. These problems depart from classical kidney exchange formulations and can be interpreted from multiple perspectives.

3.1 OVERVIEW OF THE MODEL

We begin by reviewing a model which is commonly used in kidney exchange literature. In the model, each node represents a *patient-donor pair*, and each directed edge indicates that the donor in the source node is compatible with the patient in the target node. This results in an unweighted directed graph $G = (V, E)$. Depending on the specific problem setting (such as restrictions on the length of allowed exchange cycles), this formulation leads to different variants of graph matching problems. In general, however, the objective is to find disjoint exchange cycles that include the maximum number of patients.

Extending the model to the version where each patient is associated with multiple *alternative donors* is straightforward. In this extension, each patient, along with all his associated proxy donors, is still represented as a single node in the graph. The compatibility edges are then defined such that a directed edge from node u to node v exists if any of the donors associated with patient u is compatible with the patient in node v. This preserves the structure of the original graph as a directed graph $G = (V, E)$, but with a potentially denser set of edges due to the increased number of donor-patient compatibility possibilities. This setting is then identical to the standard patient-donor pair KE problem: the objective is still to find disjoint exchange cycles in a directed compatibility graph.

A more general scenario arises if we allow a patient to receive a kidney while all of their associated donors are permitted to donate simultaneously. We call this model *multiple donors* model. How this model relates to other kidney exchange models is shown in Figure 3.1. In this case, one of the donors may participate in an exchange cycle

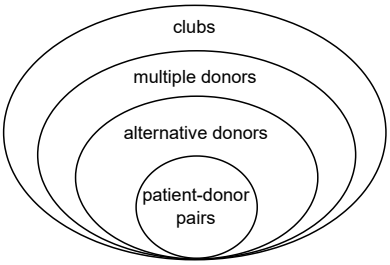


Figure 3.1: Hierarchy of kidney exchange models. Each successive set is a superset of the previous: patient-donor pairs form the core, extended by alternative donors, multiple donors, and finally clubs briefly mentioned in Section 1.1.

as usual, while the other donor initiates a donation chain, effectively increasing the total number of transplants.

In such settings, we observe that allowing chains to follow cycles can only improve social welfare—it never reduces the total number of transplants compared to *patient-donor pairs* model. Moreover, this problem deviates from the classical graph matching framework: it is no longer a pure matching problem.

Finally, we note that it is now possible for one donor to participate in a cycle while the second donor from the same patient initiates or joins a chain. This creates a structural asymmetry in the problem. A general graph representation (i.e. grouping patient and donor into one node) is no longer sufficient. Instead, we need to explicitly distinguish between patients and donors in the model to accurately capture the structure and constraints of the problem. An example of a graph with a patient having two proxy donors is shown in Figure 3.2.

The *multiple donors* model is a generalization of the *alternative donors* model and therefore also accommodates groups consisting of a patient with alternative proxy donors. This case is handled in the same way as in the *alternative donors* model—the donors are merged into a single node, as shown in Figure 3.3.

Throughout the thesis, we will use the *multiple-donors* model—that is, the one that distinguishes between node types. We also restrict the model to the case where each patient has at most two proxy donors. In the diagrams, patients will be represented as circles and donors as squares. Compatibility edges will be represented as solid arrows and proxy edges as dotted arrows.

3.2 FORMAL DEFINITIONS

We formalize the *multiple donors* model as a directed graph $G = (V, E)$, where the vertex set $V = P \cup D$ consists of two disjoint subsets: a set

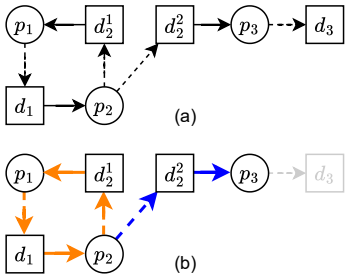


Figure 3.2: **(a)** is an example of a graph where a patient has two proxy donors. **(b)** illustrates that this allows the model to select the cycle $(p_2 \to d_2^1 \to p_1 \to d_1 \to p_2)$ (in orange) and the chain $(p_2 \to d_2^2 \to p_3)$ (in blue).

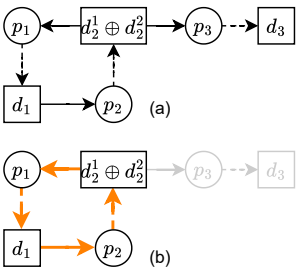


Figure 3.3: **(a)** is an example of a graph where a patient has two alternative proxy donors. **(b)** shows that only the cycle $(p_2 \to d_2^1 \to p_1 \to d_1 \to p_2)$ is selected in the solution.

of patients P and a set of donors D. The edge set E is partitioned into two disjoint subsets, $E = E_{\text{proxy}} \cup E_{\text{compatible}}$, defined as follows:

- The set of *proxy edges* $E_{\text{proxy}} \subseteq P \times D$ contains an edge (p, d) if donor $d \in D$ is a proxy donor of patient $p \in P$.
- The set of *compatibility edges* $E_{\text{compatible}} \subseteq D \times P$ contains an edge (d, p) if donor $d \in D$ is compatible with patient $p \in P$.

Each donor is a proxy for exactly one patient, and each patient has at most two proxy donors. We assume that no donor is compatible with their own associated patient, i.e., for all $(p, d) \in E_{\text{proxy}}$, we have $(d, p) \notin E_{\text{compatible}}$.

A solution is a subgraph $G^* = (V, E^*)$ with $E^* \subseteq E$ satisfying the following constraints:

1. **Patient receives at most one kidney:** Each patient receives at most one kidney from a compatible donor:

$$\forall p \in P, \quad \left|\{d \in D : (d, p) \in E^*_{\text{compatible}}\}\right| \leqslant 1.$$

2. **Donor donates only if proxy patient is matched:** A donor d may donate a kidney for some patient p if and only if his associated patient p' also receives a compatible kidney, i.e.,

$$(d, p) \in E^*_{\text{compatible}} \iff (p', d) \in E_{\text{proxy}} \text{ and } \exists (d', p') \in E^*_{\text{compatible}}.$$

As mentioned in Chapter 2, our goal is to design algorithms for this model that ideally satisfy the following three properties: (i) truthfulness (DSIC), (ii) maximization of social welfare, and (iii) computational efficiency.

3.3 PROBLEM VARIANTS

The model introduced above captures the essential structure of multi-donor kidney exchange and donation. We now build upon this model to define specific algorithmic problems under various settings. The remainder of this paper will focus on analyzing and solving these problems.

As a starting point, consider the most general setting, where no restrictions are imposed on the length of exchange cycles or donation chains. Given a directed graph $G = (V = P \cup D, E)$, all subsets of edges that satisfy the feasibility constraints described earlier are considered valid solutions. We refer to this formulation as the *Unbounded Multi-Donor Exchange Problem*.

We then consider a more constrained version, where limits are placed on the maximum allowed lengths of cycles and chains. Specifically, in the n-*Cycle* m-*Chain Problem*, only subsets in which all cycles have length at most n and all chains have length at most m are considered valid. Here, the length refers to the number of compatible edges the cycle or the chain contains. Returning to the example from Figure 3.2, the cycle has length 2 and the chain has length 1.

3.4 ALTERNATIVE INTERPRETATIONS

The problem we consider has a structured, graph-like nature. This subsection explores how our model connects to classical problem frameworks, providing alternative perspectives that inform both algorithmic design and empirical evaluation.

3.4.1 Weighted Set Packing

One useful interpretation frames our problem as a special case of the *weighted set packing* problem. Each feasible subset corresponds to a valid exchange (either a cycle or a cycle followed by chain(s)) with the constraint that no individual (patient or donor) can appear in more than one subset. The objective is to select a collection of disjoint subsets that maximizes the total number of patients matched, which defines the weight of each subset. We adopt this perspective in the design of a randomized mechanism presented in Chapter 6.

3.4.2 Integer Linear Program Formulation

Additionally, our problem can be formulated as an *integer linear program (ILP)*. In the experimental section Chapter 7, we implement and solve such an ILP to evaluate the practical effectiveness and scalability of our approach.

Let:

- P be the set of all patients.
- D be the set of all donors.
- $\mathtt{Cycles}$ be a set of all possible 2-cycles formed by patients and donors from P and D.
- $\mathtt{Chains}$ be a set of all possible chains of length 1 starting from a cycle.

The ILP formulation of *2-cycle 1-chain kidney exchange problem* is given as:

$$
\begin{aligned}
\max \quad & \sum_{c \in \text{Cycles}} 2x_c + \sum_{e \in \text{Chains}} x_e \\
\text{s.t.} \quad (1) \quad & \sum_{\substack{c \in \text{Cycles} \\ d \in c}} x_c + \sum_{\substack{e \in \text{Chains} \\ e.\text{start}=d}} x_e \leqslant 1 && \forall d \in D \\
(2) \quad & \sum_{\substack{c \in \text{Cycles} \\ p \in c}} x_c + \sum_{\substack{e \in \text{Chains} \\ e.\text{end}=p}} x_e \leqslant 1 && \forall p \in P \\
(3) \quad & \sum_{\substack{c \in \text{Cycles} \\ p_i \in c}} x_c \geqslant x_e && \forall e = (d_i, p_j) \\
& x_c \in \{0,1\} && \forall c \in \text{Cycles} \\
& x_e \in \{0,1\} && \forall e \in \text{Chains}
\end{aligned}
$$

(1) and (2) ensure that every donor and patient is selected in at most one cycle or chain. (3) ensures that a chain is started from a donor whose proxy patient is already selected in a cycle. There, (p_i, d_i) is a proxy edge and (d_i, p_j) is a compatible edge. This also ensures that the length of the chain is 1. The objective is to select a set of cycles and chains that maximize the number of matched patients.

4 INCENTIVES AND TRUTHFULNESS

4.1 MOTIVATING EXAMPLE: INCENTIVES TO REPORT MULTIPLE DONORS

Let's begin by defining the incentives of each patient-donor group. Naturally, the primary goal of each group is to obtain a compatible kidney for the associated patient. Secondarily, each group aims to minimize its loss, that is, the number of proxy kidneys it gives up in exchange for a compatible kidney. This leads to the following ranking of possible outcomes for each group, ordered from most to least desirable:

1. Receive one compatible kidney and lose 0 proxy kidneys.
2. Receive one compatible kidney and lose 1 proxy kidney.
3. Receive one compatible kidney and lose 2 proxy kidneys.
4. Receive no compatible kidney and lose 0 proxy kidneys.

Note that we do not consider a case where a group receives no compatible kidney but still loses a proxy kidney. If that were the outcome for some group, the group would simply refuse to participate. Furthermore, if a group chooses to participate as an alternative-donors group, the third outcome is not taken as an option for this group.

It is not immediately obvious whether participating in the *multiple-donors* program provides any advantage for a patient-donor group compared to participating in the *alternative-donors* program. The example in Figure 4.1 clarifies this question. In **(a)**, we see an example graph where patient p_1 has two proxy donors: d_1^1 and d_1^2. The graph includes two overlapping cycles: one of length two that includes p_1, and another of length three that does not. In the solution, only one of these cycles can be selected.

If patient p_1 chooses to report their donors as alternative donors, $d_1^1 \oplus d_1^2$, then the system (which maximizes the number of transplants) will select the longer cycle (the orange one in **(b)**), resulting in p_1 not receiving a compatible kidney. On the other hand, if p_1 is willing to use both proxy donors, then (as shown in **(c)**) the system will select the cycle of length two that includes p_1 and the chain initiated by d_1^2, also of length two, saving a total of four patients.

This scenario is not far removed from real life. Imagine that p_1 is a child in need of a kidney transplant, and d_1^1 and d_1^2 are the child's

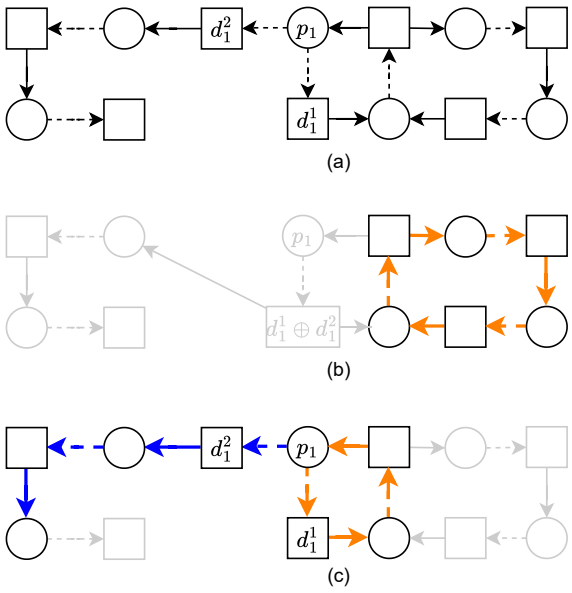


Figure 4.1: **(a)** shows an example of a graph containing two overlapping cycles, where the left cycle contains patient p_1 who has two proxy donors: d_1^1 and d_1^2. In **(b)**, patient p_1 reports his donors as alternative donors $d_1^1 \oplus d_1^2$ and as a result doesn't receive a compatible kidney. **(c)** shows that by reporting two proxy donors in the multiple-donors model, patient p_1 receives a compatible kidney in exchange for two proxy kidneys.

parents, both incompatible with the child but willing to do anything to help. The *multiple-donors* model opens new possibilities for such groups in these situations.

4.2 TRUTHFULNESS IN THE MULTIPLE-DONORS MODEL

Unfortunately, the *multiple-donors* model, together with the incentives discussed in Section 4.1, forces patients to act strategically. The question every patient will ask themselves is whether they should report all donors. If they report two donors, they risk both donors losing their kidneys, while in reality, reporting only one might be sufficient for the patient to receive a kidney. We consider two types of truthfulness.

Definition 1 (Truthfulness)**.** We say a kidney exchange mechanism is *truthful* if, for each group, donating more proxy kidneys implies receiving strictly more compatible kidneys. In other words, to satisfy truthfulness, the system should take two proxy kidneys from a group

if and only if taking one of the proxy kidneys would not result in the patient of the group receiving a compatible kidney.

Definition 2 (Weak Truthfulness)**.** We say a kidney exchange mechanism is *weakly truthful* if, for each group, reporting more proxy donors implies receiving at least as many compatible kidneys. That is, if a group reports only one proxy donor and receives a compatible kidney, then the same group would also receive a compatible kidney if it reported more proxy donors.

Of course, truthfulness implies weak truthfulness, but not vice versa.

4.2.1 Truthfulness

Lemma 1. *The social-welfare maximizing multiple donor model is not truthful.*

Proof. Let there be an algorithm A solving *social-welfare maximizing multiple donors* problem and consider the example graph from Figure 3.2. In the example, if patient p_2 reports both of his proxies; d_2^1 and d_2^2, A assigns him a compatible kidney from d_1 in exchange for two proxy donors. On the other hand, if patient p_1 decides to hide donor d_2^2, A assigns him a kidney from d_1 in exchange for only one proxy kidney d_2^1. □

Moreover, knowing the structure of the graph in the social-welfare maximizing problem gives players an advantage, which is perfectly depicted in Figure 4.2, which shows two patients being in the Prisoner's Dilemma situation. In the example, two patients p_1 and p_2 have two proxy donors; d_1^1, d_1^2 and d_2^1, d_2^2 respectively. Both patients are in the same cycle of length 3, which overlaps with a larger one of length 4. If both decide to only report one of their donors (as shown in **(b)**), the larger cycle is selected in the solution, and both patients end up without a compatible kidney. If both decide to report all of theirs donors (see **(c)**), the smaller cycle is selected together with two chains, each of length 2, summing up to 7 patients saved. In this case, both patients receive a compatible kidney, but at the same time, both lose two of theirs proxy donors. If patient p_1 knows the structure of the graph and knows that p_2 will report two of his donors, then, he can decide to hide donor d_1^2 (see **(d)**). In that case, the smaller cycle is selected together with one chain of length 2 initiated by d_2^2, summing up to total of 5 patients saved. Notably, patient p_1 received a compatible kidney and spent only one of his donors, while, p_2 spent two of his proxies.

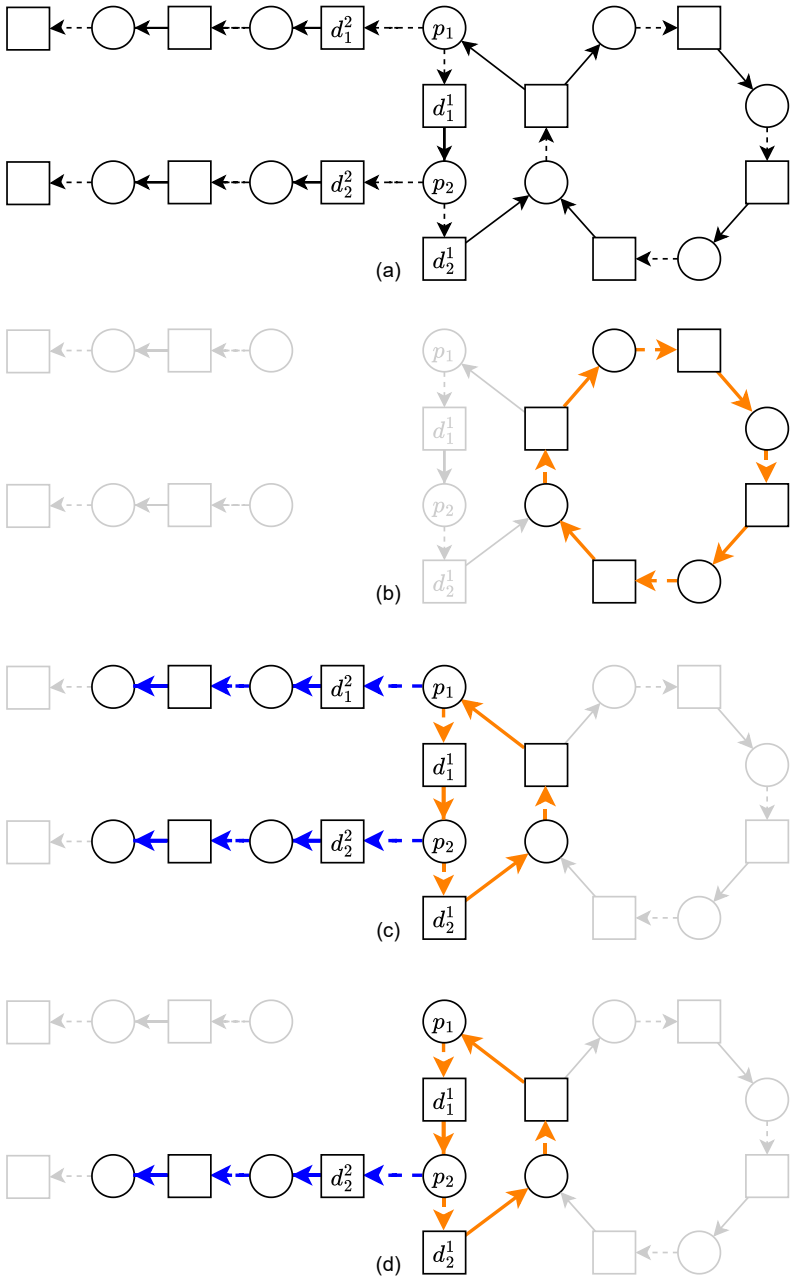


Figure 4.2: **(a)** illustrates a graph where patients p_1 and p_2 are in *Prisoner's Dilemma* situation. **(b)** shows that if both p_1 and p_2 decide not to report d_1^2 and d_2^2 respectively, they both don't receive a compatible kidney in the final solution. In **(c)**, both patients reported both of their donors; they both receive compatible kidneys, but also they both spent two kidneys from their proxy donors. In **(d)**, patient p_2 reported both of his proxy donors, patient p_1 reported only d_1^1. In this case, p_1 gets a compatible kidney and spends only one proxy kidney. Patient p_2 gets a compatible kidney as well, but spends two proxy kidneys. The symmetric case occurs when p_1 reports two proxy donors and p_2 reports only one of his proxies.

4.2.2 Weak Truthfulness

When it comes to weak truthfulness, not every algorithm that solves the social-welfare maximizing multiple donor problem possesses this property. This is demonstrated in the following lemma.

Lemma 2. *Not every algorithm solving the social-welfare maximizing multiple donors problem has weak truthfulness property.*

Proof. Consider, for example, the simple graph illustrated in Figure 4.3, and let us examine an algorithm A that solves the model without a consistent tie-breaking rule. That is, if there are multiple optimal solutions, the algorithm deterministically[1] returns one of them arbitrarily.

In this example, patient-donor groups form two overlapping cycles, each of length two. The algorithm can select only one of these cycles in its solution. Suppose A selects the left cycle, as shown in **(a)**, thereby assigning a compatible kidney to patient p_1. In **(b)**, the same patient p_1 reports an additional proxy donor d_1^2, who is not compatible with any patient in the graph. Although this addition does not affect the number of patients that can be saved, it changes the structure of the graph and may cause A to select the right cycle instead, which does not include p_1. □

The proof of the previous lemma shows that inconsistent tie-breaking is the reason an algorithm may fail to be weakly truthful. In the next lemma, we show that an algorithm with an arbitrary consistent tie-breaking rule is weakly truthful. Before we do that, let's define what consistent tie-breaking even means.

Definition 3 (Consistent tie-breaking rule)**.** Given the set S_k of all solutions that achieve the same score k (i.e., the same number of patients saved), a consistent tie-breaking rule imposes a strict, arbitrary ordering over S_k that is independent of the reports provided by patient-donor groups. For any input graph G where the maximum number of patients that can be saved is k, the rule selects the highest-ranked valid solution from S_k according to this fixed ordering.

A particularly appealing, consistent tie-breaking rule is based on assigning priorities among the patients, similar to how waiting lists for kidneys from deceased donors are handled. In such lists, patients are ranked according to a fixed priority score that reflects medical urgency, ethical considerations, or logistical factors. When a kidney becomes available, it is allocated to the highest-priority patient among those compatible with the organ. Translating this idea to our mechanism design setting, we can impose a consistent tie-breaking rule by predefining a strict priority ordering over the agents. When multiple

1 Here, *deterministically* means that given the same graph twice, the algorithm will always return the same solution for both instances.



Figure 4.3: Example outputs from an algorithm solving the social-welfare-maximizing multiple donors model without a consistent tie-breaking strategy. In **(a)**, the algorithm selects the left cycle of length 2, assigning p_1 a compatible kidney. In **(b)**, patient p_1 reports an additional 'dangling' donor d_1^2. This time, the algorithm selects the right cycle of length 2, which is equally compelling, resulting in patient p_1 not being matched to a compatible kidney.

allocations yield the same utility, the tie is broken in favor of the allocation that benefits the agent with the highest priority.

Lemma 3. *An algorithm with a consistent tie-breaking rule solving the social-welfare maximizing multiple donors model is weakly truthful.*

Proof. Let A be an algorithm with a consistent tie-breaking rule (as defined in Definition 3) that solves the social-welfare maximizing multiple donors problem, and consider an arbitrary graph G in which patient p_1 reports one proxy donor d_1^1. Assume that, given G, A returns a solution s in which p_1 receives a compatible kidney.

Now, construct a graph G' with the same structure as G, but where p_1 has an additional proxy donor d_1^2. If this addition increases the maximum number of patients saved, it implies that d_1^2 is used, and therefore p_1 still receives a compatible kidney.

If the addition of d_1^2 does not increase the number of patients saved (i.e., the optimal value remains the same), then the solution returned by A in G' must still belong to the set S_k of solutions that save k patients. Since A uses a consistent tie-breaking rule and the ordering over S_k is fixed independently of the reports, the algorithm will either:

1. Return the same solution s as before, in which p_1 receives a compatible kidney; or

2. Return a different solution $s' \in S_k$ that is ranked higher than s according to the consistent ordering. This change can only occur if d_1^2 is used, which again implies that p_1 receives a compatible kidney.

□

4.3 ACHIEVING TRUTHFULNESS VIA RANDOMIZED MECHANISMS

As discussed earlier, no deterministic algorithm can simultaneously guarantee both truthfulness and maximum social welfare in our setting. The core issue is that if a patient can obtain a kidney by reporting only one donor, they have no incentive to report both; the additional donor does not improve their utility. This misalignment can be mitigated by introducing randomness into the mechanism.

The key idea is to shift the notion of utility: instead of a binary outcome (receiving a kidney or not), we consider the probability of receiving a kidney as the utility. This enables a simple incentive structure: the more donors a patient truthfully reports, the higher their probability of being matched.

Suppose we have an algorithm A that solves the maximum social welfare problem, which is not truthful but weakly truthful. We can construct a randomized algorithm R that is truthful, using the following procedure:

1. Run the original algorithm A and obtain its output.
2. Let the output include n patients, and suppose these patients have collectively reported m donors. Let $\epsilon > 0$ be a small constant.
3. Return the output of A with probability $1 - (2n - m)\epsilon$; otherwise (with probability $(2n - m)\epsilon$), return an empty allocation.

The expected performance of R satisfies:

$$\lim_{\epsilon \to 0^+} \frac{\mathbb{E}[R]}{O} = 1 - (2n - m)\epsilon = 1,$$

implying that the randomized algorithm R achieves nearly the same performance as O when ϵ is small.

The underlying intuition is straightforward: the key factor for incentive alignment is whether a patient receives a kidney. In the randomized mechanism, patients are incentivized to report as many valid donors as possible. The more donors they report, the smaller the quantity $(2n - m)$ becomes, thereby reducing the probability that the output is discarded and increasing their expected utility.

Lemma 4. *Let A be an algorithm that is weakly truthful. Then the randomized mechanism R constructed as described is truthful.*

Proof. (Proof sketch) To show that the randomized mechanism R is truthful, consider a report profile s in which all patients report truthfully, and let patient i report both of their available donors. Let s_{-i} denote a modified report where patient i misreports by reporting only one of the two donors.

Suppose first that, under s_{-i}, patient i receives a kidney in the output of algorithm A. Since A is weakly truthful, patient i would also receive a kidney under s. Therefore, in the randomized mechanism R, the probability that patient i receives a kidney under s is strictly higher than under s_{-i}, since the value $(2n - m)$ is smaller when both donors are reported. That is, $R(s)$ has a higher expected utility than $R(s_{-i})$.

Now suppose that under s_{-i}, patient i does not receive a kidney in A. In that case, reporting both donors cannot result in a worse outcome under R, and may result in a better one. In both scenarios, truthful reporting yields a weakly higher expected utility.

Thus, for every patient, reporting all available donors is a dominant strategy in the randomized mechanism R, and therefore R is truthful. □

5 COMPUTATIONAL COMPLEXITY ANALYSIS

This chapter investigates the computational complexity of the algorithmic problems introduced in our multi-donor kidney exchange model. We formalize the decision versions of these problems and show that they are NP-complete, even under seemingly simple constraints.

We begin by presenting several variants of the problem that differ in the allowed structure of exchanges, such as the maximum length of cycles and chains. We then prove that all of these variants are NP-complete. These results motivate the need for approximation algorithms and heuristic methods, which will be discussed in later chapters.

5.1 PROBLEM VARIANTS AND OVERVIEW OF RESULTS

All problem variants considered in this chapter revolve around structural constraints imposed on the kidney exchange process, specifically, limitations on the maximum length of allowed exchange cycles and chains.

We begin by showing that for any fixed constant $k \geqslant 2$, the decision version of the k-*Cycle Unbounded-Chain Multi-Donor Kidney Exchange Problem* is NP-complete. This result is established via a reduction from the *Longest Path from Fixed Source* problem.

Subsequently, we show that for all $n \in \{n : n \geqslant 2\} \cup \{\infty\}$ and $m \in \{m : m \geqslant 1\} \cup \{\infty\}$, *n-Cycle m-Chain Multi-Donor Kidney Exchange Problem* is computationally hard, using a reduction from the *2P2N 3-SAT* problem.

5.2 REDUCTION FROM LONGEST PATH

In this section, we formally establish the computational hardness of the matching problems defined in our multi-donor kidney exchange model. Specifically, we show that even under simple structural constraints (such as allowing only one 2-cycle and unrestricted chains), the decision version of the problem remains NP-complete.

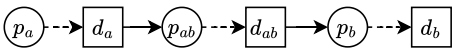


Figure 5.1: Edge gadget used in reduction from Longest Path from Fixed Source problem.

5.2.1 NP-Completeness of the 2-Cycle Unbounded-Chain Problem

Lemma 5. *The decision version of the 2-Cycle Unbounded-Chain Multi-Donor Kidney Exchange Problem is NP-complete.*

Proof Sketch. We prove NP-hardness by reduction from the *Longest Path from Fixed Source* problem (Problem 1). Given a directed graph $G = (V, E)$, a designated source node s, and an integer k, we construct an instance of the *2-Cycle Unbounded-Chain Multi-Donor Kidney Exchange* problem such that there exists a solution saving at least $3 + 2k$ patients if and only if there exists a simple path of length at least k starting from s in G.

To construct the kidney exchange instance, we begin by adding a single 2-cycle between two patient-donor groups, where one of the patients is associated with two donors: one donor participates in the 2-cycle, and the other donor is compatible with the patient corresponding to node s in G. This setup initiates the only possible chain in the instance.

Next, for each vertex $v \in V$, we create a corresponding patient-donor pair; p_v, d_v. For every directed edge $(a, b) \in E$, we introduce a new intermediate patient-donor pair; p_{ab}, d_{ab}, and add two compatibility edges; (d_a, p_{ab}) and $(d_a b, p_b)$. This models each edge as a two-step donation chain segment, as shown in Figure 5.1.

In the resulting kidney exchange instance, only the single 2-cycle we constructed initially is eligible to be selected under the problem constraints. Chains can only be initiated from this 2-cycle and must respect donor compatibility. Since all other potential cycles in the graph (formed via edge constructions) involve at least four patients, no other 2-cycles exist in the instance.

Therefore, maximizing the number of patients who receive a kidney corresponds to finding the longest simple path in G starting from s. Each edge in G contributes two transplantations (through the intermediate patient-donor pair), and the initial 2-cycle saves two patients, plus one more as the entry to the chain. Hence, the total number of patients saved is at least $3 + 2k$ if and only if there exists a simple path of length at least k in the original graph.

An example instance of *Longest Path from Fixed Source* problem and its corresponding reduction to the instance of the *2-Cycle Unbounded-Chain Multi-Donor Kidney Exchange* problem is shown in Figure 5.2.

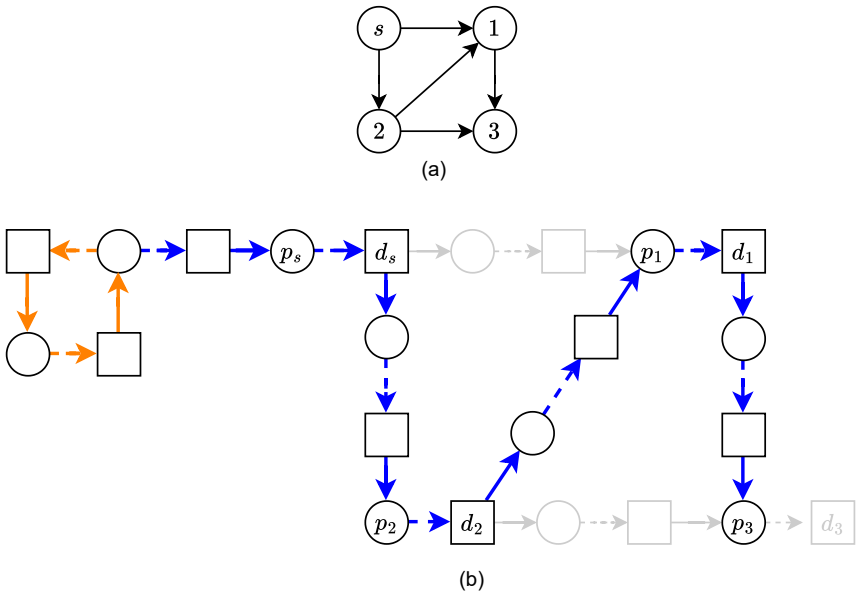


Figure 5.2: **(a)** shows an example graph instance of *Longest Path from Fixed Source* problem, where the longest path $(s \to 2 \to 1 \to 3)$ has length 3. **(b)** illustrates the same graph represented in the *2-Cycle Unbounded-Chain Multi-Donor Kidney Exchange* problem and the corresponding optimal solution. The orange 2-cycle on the left initiates the longest available chain consisting of 7 transplantations. The solution sums up to 9 patients saved in total.

Membership in NP is immediate, since a proposed matching (i.e., a set of selected cycles and chains) can be verified in polynomial time. □

5.2.2 Extensions to n-Cycle Constraints

We now extend the previous hardness result to two broader settings, where the constraint on the length of cycles is generalized from length 2 to an arbitrary constant $n \geqslant 2$.

Lemma 6. *For any fixed constant $n \geqslant 2$, the decision version of the n-Cycle Unbounded-Chain Multi-Donor Kidney Exchange Problem is NP-complete.*

Proof Sketch. The proof follows a similar reduction as in the 2-cycle case. At the beginning, we construct a cycle of length 2 that initiates a chain.

We ensure that this is the only valid cycle in the constructed. To achieve this, we modify the edge gadgets from the original reduction. Each edge (a, b) is now represented by an expanded chain segment, composed of additional n intermediate patient-donor pairs, such that any cycles formed from these components are of length strictly greater than n. This guarantees that under the n-Cycle constraint, the only valid cycle is the one we intentionally constructed for initiating the chain.

Thus, as in the 2-cycle case, any feasible solution must rely on a single chain starting from the constructed n-cycle. The total number of patients can be served as a function of the chain's length, which in turn corresponds to the length of a simple path in the original graph. Therefore, the reduction remains valid for arbitrary fixed n. □

5.3 REDUCTION FROM 2P2N-3SAT

In the following two lemmas, we show that for all $n \in \{n : n \geqslant 2\} \cup \{\infty\}$ and $m \in \{m : m \geqslant 1\} \cup \{\infty\}$, *n-Cycle m-Chain Multi-Donor Kidney Exchange Problem* is computationally hard. This result is established via a reduction from the *2P2N-3-SAT* problem (Problem 2).

Lemma 7. *The decision version of the 2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem is NP-complete.*

Proof Sketch. We prove NP-hardness via a reduction from the *2P2N-3-SAT* problem. Let ϕ be a Boolean formula in conjunctive normal form, consisting of a set of clauses C and a set of variables X, where each clause $c \in C$ contains exactly three literals, and each variable $x \in X$ appears in two clauses as a positive literal x and in two clauses as a negative literal $\neg x$. We construct a corresponding instance of the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem* such that ϕ is

satisfiable if and only if there exists a feasible exchange satisfying a certain number of patients t (specified below).

In the construction, each clause $c \in C$ is represented as a patient-donor pair (p_c, d_c). For each variable $x \in X$, we construct a special gadget illustrated in Figure 5.3. The gadget consists of patient-donor groups forming four overlapping cycles and four patient-donor pairs representing two negative literals ($\neg x$) and two positive literals (x). The gadget is designed so that either the two patients representing the positive literals p_x are satisfied (as shown in **(b)**), or the two patients representing the negative literals $p_{\neg x}$ are satisfied (as shown in **(c)**), but never both p_x and $p_{\neg x}$ simultaneously. The patients representing clauses and those representing literals are the same nodes in our construction. For example, given a clause $c = (x_1, \neg x_2, x_3)$, we identify $p_c = p_{x_1} = p_{\neg x_2} = p_{x_3}$.

In Figure 5.3, notice that no matter which pair of cycles is selected, the four patients p_1, p_2, p_3, and p_4 always receive compatible kidneys. If ϕ is satisfiable, then in our construction, all patients representing clauses $p_c : c \in C$ receive compatible kidneys in every optimal solution. Conversely, if ϕ is not satisfiable, then in every solution, there exists at least one patient representing a clause who does not receive a compatible kidney. This gives rise to the threshold $t = 4 \cdot |X| + |C|$, indicating that the solution to the constructed instance of the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem* satisfies t patients if and only if ϕ is satisfiable.

The problem is clearly in NP. Given a proposed solution to the instance of the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem*, we can verify in polynomial time whether it satisfies at least $t = 4 \cdot |X| + |C|$ patients.

Since the problem is both in NP and NP-hard, we conclude that the decision version of the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem* is NP-complete. □

Lemma 8. *For all* $n \in \{n : n \geqslant 2\} \cup \{\infty\}$ *and* $m \in \{m : m \geqslant 2\} \cup \{\infty\}$, *The decision version of the n-Cycle m-Chain Multi-Donor Kidney Exchange Problem is NP-complete.*

Proof Sketch. This proof is almost identical to the one of Lemma 7. What changes is the structure of the variable gadget (in Figure 5.4) and the threshold t. This time the variable gadget consists of two overlapping cycles of length two, which only one of which can be selected. Each of the cycles, when selected, start a donation chain of length two, that can satisfied up to two patients representing positive literals p_x or up to two patients representing negative literals $p_{\neg x}$, as it is shown in **(a)** and **(b)**. Notice, that in this construction, there are no other cycles and no other chains than the ones just mentioned. That implies, the gadget works for arbitrary values of $n \in \{n : n \geqslant 2\} \cup \{\infty\}$

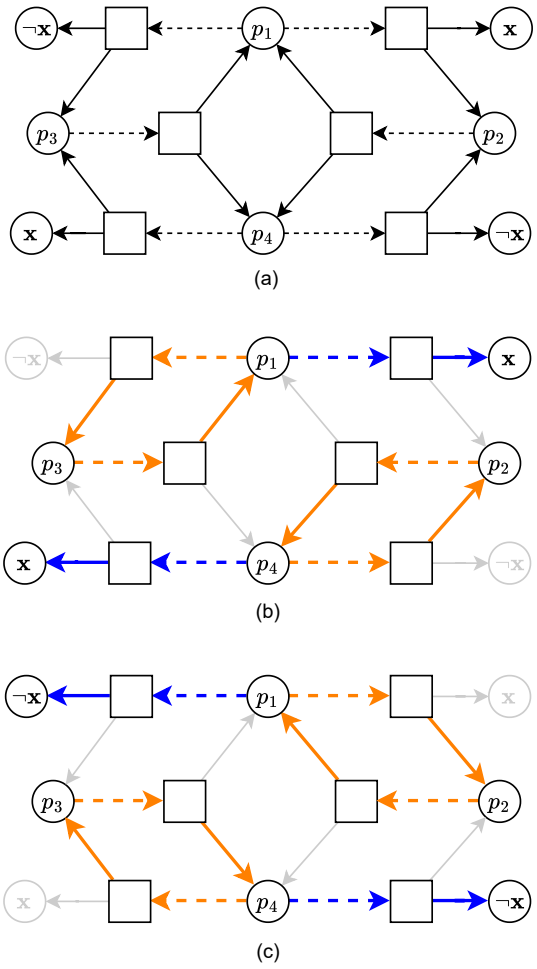


Figure 5.3: Boolean variable gadget used in the reduction from *2P2N 3-SAT* to the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem*. **(a)** The full variable gadget, representing variable x. **(b)** An assignment where the variable is set to `true`, **(c)** An assignment where the variable is set to `false`. Only one of these two assignments can be selected in a valid matching, ensuring that either x or ¬x is selected, but not both.

and $m \in \{m : m \geqslant 2\} \cup \{\infty\}$, since only 2-cycles and chains of length 2 are required to encode the variable gadget and satisfy the literals.

The clause gadgets remain unchanged from the previous proof. As before, each clause $c \in C$ is represented by a patient-donor pair (p_c, d_c). Thus, a clause is satisfied in the exchange if and only if it gets a kidney from one of the associated variable gadgets.

The correctness of the reduction follows similarly: if the Boolean formula ϕ is satisfiable, then there exists a consistent assignment of truth values to the variables such that, for each variable x, we select the cycle corresponding to either the positive or the negative assignment. This initiates chains satisfying the corresponding literals (and in turn the clauses, since a clause and its corresponding literals are represented by the same patient). As a result, all patients corresponding to clauses receive compatible kidneys, reaching the threshold $t = 3 \cdot |X| + |C|$.

Conversely, if ϕ is not satisfiable, then for every selection of variable gadgets, at least one clause cannot be satisfied—meaning its corresponding patient does not receive a kidney. Therefore, the threshold t cannot be reached, and no feasible exchange meets the requirement.

Hence, the decision version of the *n-Cycle m-Chain Multi-Donor Kidney Exchange Problem* is NP-complete for all $n \geqslant 2$ and $m \geqslant 2$, including the unbounded cases.

□

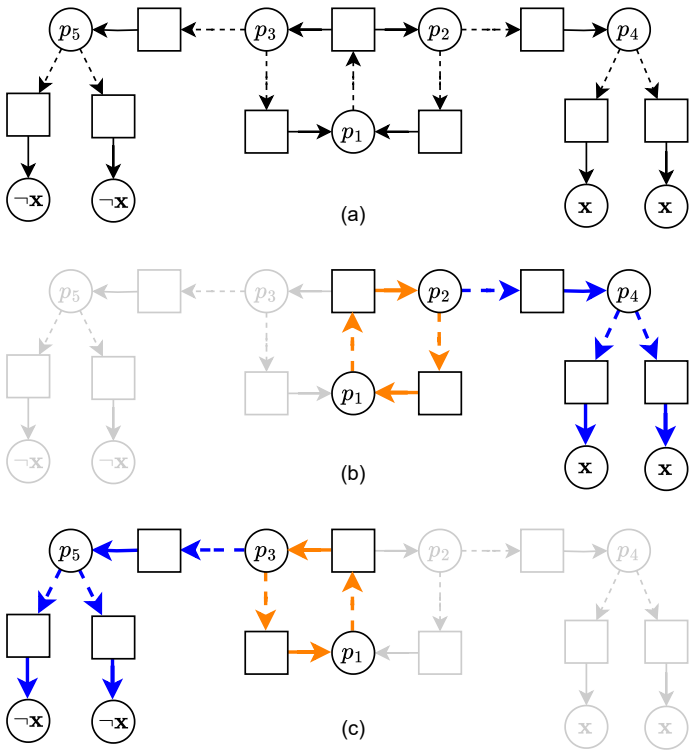


Figure 5.4: Boolean variable gadget used in the reduction from *2P2N 3-SAT* to the *n-Cycle m-Chain Multi-Donor Kidney Exchange Problem*. **(a)** The full variable gadget, representing variable x. **(b)** and **(c)** are assignments where the variable is set to `true` and `false` respectively.

6 APPROXIMATION ALGORITHMS AND INAPPROXIMABILITY

In this chapter, we investigate the approximability of the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem*. We first present a simple approximation algorithm that leverages known results from matching and set packing. We then analyze its approximation ratio and demonstrate that it achieves a constant strictly better than 0.5.

Based on the two reductions in Chapter 5, we can also get some results for the hardness of approximation.

6.1 APPROXIMATION ALGORITHM FOR MULTI-DONOR EXCHANGE

A natural baseline heuristic is to select only disjoint 2-cycles and ignore all chains. This corresponds to the *alternative donors* model described in Chapter 3, where each patient has one or more donors, but only 2-cycles are allowed. The problem then reduces to maximum matching in an undirected graph, solvable in polynomial time via Edmonds' Blossom algorithm [5]. Since each 2-cycle can potentially support up to two chain transplants, this strategy yields a 0.5-approximation in the worst case.

To improve upon the baseline, we reinterpret our problem as a special case of the *weighted* k*-Set Packing problem*. Details in Section 3.4.1.

We now describe a hybrid algorithm that combines multiple such approximations to achieve a provable bound strictly better than 0.5. We mentioned a polynomial time algorithm solving Problem 3, achieving $\frac{3}{k+1+\varepsilon}$ approximation ratio for any $\varepsilon > 0$. Based on this, we introduce the following algorithm.

Lemma 9. *The following algorithm achieves an approximation ratio of at least* $\frac{36}{67} - \varepsilon$ *for any* $\varepsilon > 0$ *for the 2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem.*

Proof. To simplify, we assume that ε is close to 0. The algorithm considers four steps:

1. Use only 2-cycles and apply the Blossom algorithm for exact maximum matching.

2. Use only 2-cycles with one additional chain (3-element subsets), and apply a $\frac{3}{4}$-approximation algorithm for unweighted 3-Set Packing.

3. Use only 2-cycles with two chains (4-element subsets), and apply a $\frac{3}{5}$-approximation algorithm for unweighted 4-Set Packing.
4. Return the best solution among the three.

Let the optimal solution contain:

- a subsets of size 4
- b subsets of size 3
- c subsets of size 2

Then:

$$\mathrm{OPT} = 4a + 3b + 2c$$

The algorithm achieves:

$$\mathrm{ALG} = \max\left(\tfrac{12}{5}a,\ \tfrac{9}{4}(a+b),\ 2(a+b+c)\right)$$

To lower-bound ALG in terms of OPT, we introduce weights $t_1, t_2 \geqslant 0$ with $t_1 + t_2 \leqslant 1$, such that:

$$\begin{aligned}\mathrm{ALG} &\geqslant \tfrac{12}{5}t_1 a + \tfrac{9}{4}t_2(a+b) + 2(1-t_1-t_2)(a+b+c)\\ &= \left(2 + \tfrac{2}{5}t_1 + \tfrac{1}{4}t_2\right)a + \left(2 - 2t_1 + \tfrac{1}{4}t_2\right)b + (2 - 2t_1 - 2t_2)\,c\end{aligned}$$

We seek the largest constant k such that:

$$\mathrm{ALG} \geqslant k \cdot \mathrm{OPT} = 4ka + 3kb + 2kc$$

The corresponding linear program is:

$$\begin{aligned}\text{maximize}\quad & k\\ \text{subject to}\quad & 2 + \tfrac{2}{5}t_1 + \tfrac{1}{4}t_2 \geqslant 4k\\ & 2 - 2t_1 + \tfrac{1}{4}t_2 \geqslant 3k\\ & 2 - 2t_1 - 2t_2 \geqslant 2k\\ & t_1 \geqslant 0,\ t_2 \geqslant 0,\ t_1 + t_2 \leqslant 1\end{aligned}$$

Solving yields:

$$\boxed{k = \frac{36}{67} \approx 0.5373}$$

Note that the approximation algorithm for unweighted 3-Set Packing does not exactly achieve a $\frac{3}{4}$ approximation, but can get arbitrarily close to it. Consequently, our overall algorithm cannot guarantee a $\frac{36}{67}$ approximation ratio, but it can approach this bound arbitrarily closely. □

6.2 HARDNESS OF APPROXIMATION

Lemma 10. *For any fixed constant $n \geqslant 2$, the maximization version of the n-Cycle Unbounded-Chain Multi-Donor Kidney Exchange Problem cannot be polynomial time approximated within $n^{1-\varepsilon}$ for any $\varepsilon > 0$ unless $P = NP$.*

Proof Sketch. We reduce from the *Longest Path from Fixed Source* problem as shown in Chapter 5. The number of matched patients is proportional to the length of a simple path in the original graph. Therefore, any approximate solution to the exchange problem would yield a comparable approximation for the longest path problem, violating its known inapproximability bounds. □

Lemma 11. *The 2-Cycle 1-Chain Multi-Donor Kidney Exchange Problem is APX-hard.*

Proof Sketch. We reduce from the *MAX-(2P,2N)-3SAT* problem, which is known to be APX-hard. The construction is the same as the one used in the reduction in Chapter 5. □

7 EXPERIMENTS

7.1 THE RANDOM GRAPH MODEL

The random graph model we use for the experiment was inspired by the one introduced in [12]. It was designed for generating patient-donor pairs based on a fixed blood type distribution and tissue-type compatibility factor $1 - p_c = 0.8$. In our setting, we consider patient-donors groups with up to two proxy donors, so the already existing random graph model had to be modified in order to fit our requirements. We introduce a new variable; *double donor ratio* (ddr), which represents the percentage of patient-donor groups with two proxy donors. The amount of information about what the value of ddr is in the real-world KE programs is limited, [8] reports that only 7.3% of candidates registered with more than one potential donor. The following are the algorithms we use for generating random patient-donor groups (Algorithm 1) and the compatibility edges between them (Algorithm 2)

Algorithm 1 : Generate patient-donor groups with multiple donors

Input : n: number of patient-donor groups to generate, ddr: double donor ratio, p_c: tissue incompatibility factor
Output : D: n patient-donor groups

```
1  D ← ∅;
2  #double_donor_groups ← n · ddr;
3  while |D| < n do
4      p ← pick random blood type;
5      if |D| < #double_donor_groups then
6          donor_types ← pick 2 random blood types;
7      else
8          donor_types ← pick 1 random blood types;
9
10     if ∀d ∈ donor_types, p is incompatible with d or
        random() < p_c then
11         D ← D ∪ {(p, donors_types)};
12 return D
```

The graph we generate can be adapted to the model with only one donor per group by either randomly removing one of the proxy donors from each group containing two proxy donors, or by setting the parameter ddr to 0 (which is equivalent to the model described in

Algorithm 2 : Generate compatibility edges

Input : D: patient-donor groups, p_c: tissue incompatibility factor

Output : E_c: compatibility edges

```
1 E_c ← ∅;
2 foreach (p_i, donors_i) ∈ D do
3     foreach (p_j, donors_j) ∈ D where i ≠ j do
4         foreach d ∈ donors_i do
5             if d is blood type compatible with p_j and
               random() ⩾ p_c then
6                 E_c ← E_c ∪ {(p_j → d)};
7 return E_c
```

[12]). Importantly, these two methods produce models of KE programs with different blood type distributions. Notably, as ddr increases, the percentage of patients with an unfavorable blood type O also increases, as shown in Figure 7.1.

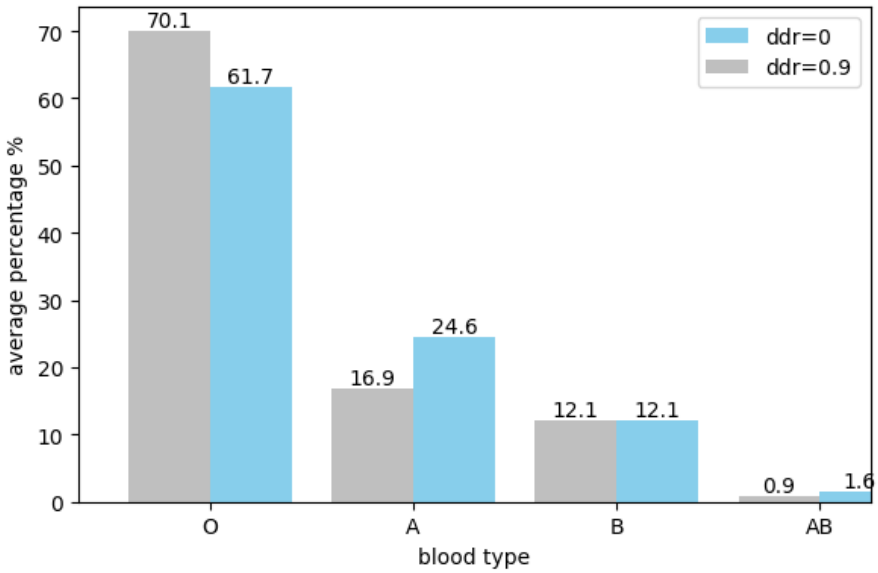


Figure 7.1: The diagram illustrating the change in the distribution of patients' blood types in KE programs as we change the ratio of groups having two proxy donors.

7.2 EXPERIMENT OVERVIEW

In this section, we present the experimental evaluation of our proposed model. We aim to investigate the practical impact of allowing multiple donors per patient under different kidney exchange settings.

Our focus is on the *2-Cycle 1-Chain Multi-Donor Kidney Exchange Model*. For comparison, we include *2-Cycle No-Chain Multi-Donor Model* (*Alternative-Donors Model*), where each patient may have up to two donors, but at most one donor can be used, and only 2-cycles are allowed.

We then formulate the resulting problem as an instance of integer programming introduced in Section 3.4.2 and solve it using IBM ILOG CPLEX Optimizer [9].

CPLEX is capable of computing optimal solutions efficiently in most cases; however, it does not guarantee short runtimes for all instances. Therefore, we set a timeout threshold: if a single task runs for more than 12 minutes, it is terminated and excluded from the results.

For the experimental analysis, we investigate two key questions.

Experiment 1. We examine how the percentage of patients matched changes as the total number of patients increases across the two models. We expect the match rates for both of them to increase slightly as the number of patients grows, due to more potential matches being available. However, we do not expect a significant increase, since the compatibility graph is already fairly dense. We expect that as the number of patients increases, the variance will decrease.

Experiment 2. We analyze how the match rate varies with the proportion of patients having a second donor. Specifically, we hypothesize the following trends:

- For the *2-Cycle No-Chain Multi-Donor Model*, the effect is ambiguous: while the increased donor pool may improve match opportunities, the changing distribution may counteract this benefit.
- For the *2-Cycle 1-Chain Multi-Donor Model*, we expect a clear increase in the match rate, as the additional donors provide more opportunities to initiate chains.

7.3 RESULTS

The following two figures (Figure 7.2 and Figure 7.3) present the experimental results. The semi-transparent region around the curve indicates the variance across runs. For each test case, we run the experiment 100 times and average the results. In the first experiment, the number of patients is fixed at 1000. In the second experiment, the double donor rate is fixed to 10%.

As expected, all observed trends are consistent with our hypotheses. In Experiment 1, the *2-Cycle 1-Chain Multi-Donor Model* achieves an improvement over the *2-Cycle No-Chain Multi-Donor Model*, increasing the match rate by 5% for all numbers of patients. In Experiment 2,

interestingly, when ddr reaches 10%, the percentage of saved patients in the *2-Cycle 1-Chain Multi-Donor Model* converges to 60%, achieving a 5% increase over the *2-Cycle No-Chain Multi-Donor Model*.

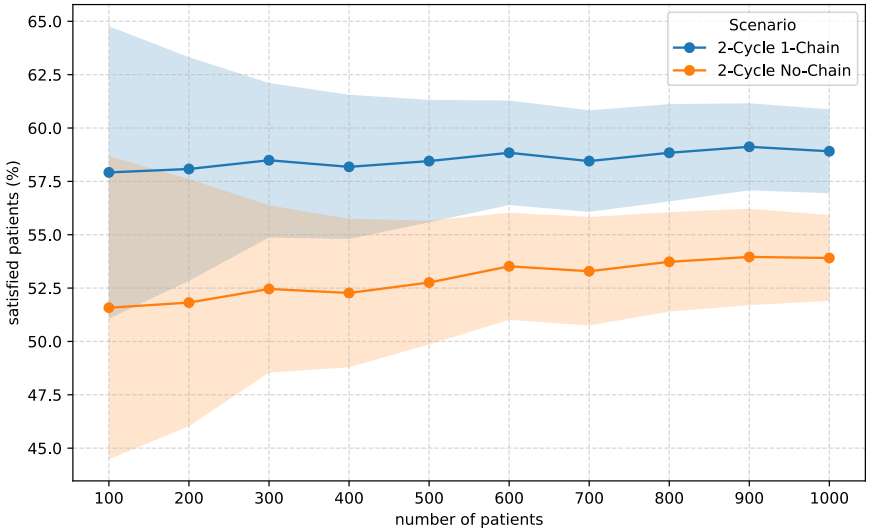


Figure 7.2: Results of Experiment 1. Average percentage of satisfied patients vs number of patients (with double donor ratio = 0.1)

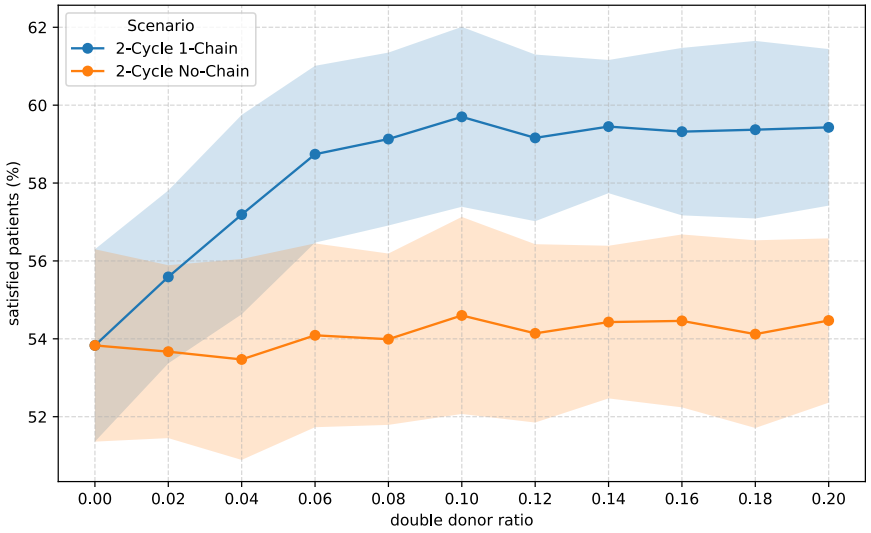


Figure 7.3: Results of Experiment 2. Average % of satisfied patients (among 1000) vs double donor ratio

Regarding computational efficiency, we observe that for most instances (up to one thousand patients), the optimal solution can be computed within one minute on a standard laptop. More specifically, out of 1,100 instances, only four exceeded the timeout threshold. This indicates that computational resources are unlikely to pose a practical bottleneck for applying our model in real-world scenarios.

8 CONCLUSION

8.1 SUMMARY

This thesis introduced and analyzed a generalized model for kidney exchange in which each patient may be associated with up to two proxy donors. We formalized this multi-donor setting into well-defined problems, capturing the structural and strategic complexities arising from allowing multiple donors per patient.

Our main contributions include a detailed computational complexity analysis, proving NP-completeness for various problem variants via reductions from classic hard problems. We also provided constant-factor approximation algorithms and established inapproximability bounds for the most relevant cases. Furthermore, we explored incentive issues and showed how randomized mechanisms can restore truthfulness. Finally, we demonstrated the practical feasibility of solving realistic instances through experiments, showing that optimal solutions can be efficiently computed and that introducing second donors significantly improves match rates.

8.2 LIMITATIONS AND OUTLOOK

While our work provides a theoretical foundation for kidney exchange with multiple donors, several limitations remain. In real-world kidney exchange programs, many additional factors must be considered. Patients are often prioritized based on factors such as waiting time or medical urgency, and compatibility edges are typically weighted to reflect the probability of a successful transplant rather than being binary. Moreover, patients often participate through hospitals or organizations, rather than individually. Finally, our model assumes a static, one-shot matching process, whereas real kidney exchange programs are dynamic and continuously evolving over time.

Therefore, future work could extend our model to incorporate the practical complexities outlined above. Additionally, in Chapter 6, our inapproximability results are primarily derived via reductions from problems in Chapter 5. Further insights may be obtained by exploring alternative techniques beyond classical reductions.

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DECLARATION

We hereby declare that this thesis is the result of our joint and independent work. The problem formulation, model development, algorithm design, experimental setup, and final analysis were all conducted by us.

During the writing process, we used OpenAI's ChatGPT to refine the wording of certain paragraphs (e.g., grammar correction and expression polishing). We also consulted the tool for common experimental utilities such as plotting scripts and basic code suggestions. However, we did not rely on AI to generate any substantive part of the thesis, nor did we use it to produce or simulate results. All core contributions reflect our own work.

We have followed academic integrity guidelines, and all sources are properly cited. We take full responsibility for the content of this thesis.

Aarhus, June 2025

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Zhang, Yuchen